Code-Layout, Readability and Reusability

Date	1 JULY 2026
Team ID	XXXXXX
Project Name	FinRelief AI – AI Powered Debt Relief & Financial Recovery Platform
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	yes	Proper and consistent indentation has been maintained throughout the project, improving code readability.
2	Proper File Structure	Files and folders are logically organized	Yes	The project follows a well-organized folder structure, separating frontend, backend, APIs, and database files.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Descriptive and meaningful variable, function, and class names are used according to coding standards.
4	Function / Method Names	Functions are descriptively named	Yes	Functions and methods are clearly named based on their functionality, making the code easy to understand.
5	Code Comments	Inline and block comments explain logic	Yes	Necessary comments have been added to explain important logic and improve maintainability.
6	Modular Design	Code is split into reusable functions/modules	Yes	The application is divided into reusable modules and components, reducing code duplication.
7	No Redundant Code	Duplicate or unused code is removed	Yes	Duplicate code has been minimized by using reusable functions and components wherever possible.

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Authentication Module	FastAPI	Login & Register	High
2	Loan management	Python	Loan Operation	High
3	Financial Engine	Python	EMI & DTI Calculation	High
4	AI Engine	Gemini API	Negotiation Letter	High
5	React Components	React.js	Dashboard UI	High
8	Error Handling	Exceptions and errors are handled gracefully	yes	Appropriate exception handling and input validation have been implemented to improve application reliability.

Reusable Components / Modules:

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5/5	The project follows a clean and well-organized folder structure with proper separation of frontend, backend, and database modules.
Readability	5/5	The code is easy to understand due to meaningful variable names, consistent formatting, and proper indentation.
Reusability	5/5	Reusable components, functions, and modules are implemented effectively, reducing code duplication and improving maintainability.
Documentation / Comments	5/5	Appropriate comments and documentation are provided for important functions, APIs, and project modules.
Overall Score	5/5	The project demonstrates excellent coding standards, modular design, maintainability, and readability. The source code is well-structured, reusable, and follows industry best practices, making it easy to understand, test, and enhance in the future.